Statistical models

Exercise Sheet 1

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Solve the following problems before the exercise class on April 17.

Problem 1
It is assumed that the mortality caused by malignant melanomas depends on the sunshine intensity, which corresponds to the latitude. The following table provides for several states in the US the number of deaths per 10 millions of the white population between 1950 and 1969 as well as the latitude.

State	Delaware	Iowa	Michigan	New Hampshire	Oklahoma	Texas	Wyoming
Mortality	200	128	117	129	182	229	134
Latitude	39	42	44	44	35	31	43

In the following, linear regression is used to explain the mortality by the latitude.

a) Estimate the intercept and the coefficient of the linear model as well as the variance of the random errors.

b) Predict the mortality for Ohio whose latitude is 40.

From now on we assume a Gaussian linear model.

c) Use a statistical test with level 0.05 to show that the slope is negative.

d) Derive a confidence interval for the mortality in Ohio with error level 0.1.

For testing hypotheses and computing confidence intervals one can use the functions `glht`, `summary.glht` and `confint.glht` from the `R` package `multcomp`. The residuals of a linear model `LM` can be accessed by `LM$residuals`.

e) Solve the problems a) to d) by using `R`.

Problem 2
In order to check the assumption of the Gaussian linear model that the error terms are i.i.d. normally distributed, one can use the following graphical tools:

- In a normal qq-plot, the empirical quantiles of the residuals are plotted against the quantiles of the standard normal distribution. If the points form approximatively a straight line, this indicates that the residuals are normally distributed. The command for a normal qq-plot in `R` is `qqnorm`.

- One can plot the residuals against the fitted values. Here, one should not see any dependence of the residuals on the fitted values.

In `R`, the residuals and fitted values of a linear model `LM` can be accessed by `LM$residuals` and `LM$fitted.values`, respectively. The command to read data from a csv-file is `read.csv`.

In the data set, `Residuals_Gaussian.csv`, which was uploaded to Stud.IP, you can find observations of Y_1, Y_2, Y_3 and U. Use `R` to compute the residuals for the three linear models where Y_1, Y_2 and Y_3 are explained by U and check with the tools described above if the random error terms are i.i.d. normally distributed.

Problem 3
The data set `Greenberg.csv`, which was uploaded to Stud.IP, provides for children the height, the information if they attended a private or a rural school and the age. One is interested in explaining the height. Use `R` for the following analyses:

a) Consider the linear model with the school as only independent variable and study whether there is a difference between the two schools. Formulate the null hypothesis and the alternative and conduct a test at level 0.05.

b) Take for the linear model the school and the age of the children into account and investigate if there is a difference between the two schools. Formulate the null hypothesis and the alternative and test at level 0.05.

Problem 4
For $n_1, n_2 \in \mathbb{N}$ with $n_1 + n_2 \geq 3$ let $\mathbf{Z}_1 = (Z_1^{(1)}, \dots, Z_1^{(n_1)})^T$ and $\mathbf{Z}_2 = (Z_2^{(1)}, \dots, Z_2^{(n_2)})^T$ with independent random variables $Z_1^{(1)}, \dots, Z_1^{(n_1)}, Z_2^{(1)}, \dots, Z_2^{(n_2)}$ such that $Z_1^{(1)}, \dots, Z_1^{(n_1)} \sim \text{Normal}(\mu_1, \sigma^2)$ and $Z_2^{(1)}, \dots, Z_2^{(n_2)} \sim \text{Normal}(\mu_2, \sigma^2)$ with unknown $\mu_1, \mu_2 \in \mathbb{R}$ and $\sigma \in (0, \infty)$. The goal is to test

$$H_0 : \mu_1 = \mu_2 \quad \text{against} \quad H_1 : \mu_1 \neq \mu_2.$$

The two-sample t-test rejects H_0 at level $\alpha \in (0, 1)$ if

$$0 \notin \left(\overline{\mathbf{Z}}_2 - \overline{\mathbf{Z}}_1 - \frac{\sqrt{n_1 + n_2}}{\sqrt{n_1 n_2}} t_{n_1+n_2-2, 1-\frac{\alpha}{2}} S, \overline{\mathbf{Z}}_2 - \overline{\mathbf{Z}}_1 + \frac{\sqrt{n_1 + n_2}}{\sqrt{n_1 n_2}} t_{n_1+n_2-2, 1-\frac{\alpha}{2}} S \right)$$

with

$$S^2 = \frac{1}{n_1 + n_2 - 2} \left(\sum_{i=1}^{n_1} (Z_1^{(i)} - \overline{\mathbf{Z}}_1)^2 + \sum_{i=1}^{n_2} (Z_2^{(i)} - \overline{\mathbf{Z}}_2)^2 \right).$$

a) Formulate the situation of the two-sample t-test as a regression problem.

b) Show that the two-sample t-test is a level-α test.